

M V S ANIRUDH, MAGANTI

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PROFESSIONAL SUMMARY

AI/ML Engineer with 7 years of experience designing and deploying machine learning and generative AI solutions across HR, finance, marketing, and manufacturing. Specialized in building LLM-powered applications (LangChain, RAG, Vector DBs), multimodal deep learning models, and production-grade ML pipelines on AWS and GCP. Proven record of delivering measurable business outcomes such as boosting churn prediction accuracy by 20% at ADP and reducing fraud investigation times by 40%. Adept at MLOps (SageMaker, MLflow, Kubeflow, CI/CD) and translating cutting-edge research into scalable enterprise solutions.

SKILL SET

- **Programming:** Python (Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Keras), SQL, R (basic), Linux.
- **AI/ML:** Supervised/Unsupervised Learning, Gradient Boosting, NLP (BERT, DistilBERT, GPT, RAG, LangChain, FAISS, Pinecone), LLM Fine-tuning, Deep Learning (CNN, U-Net, LSTM, GRU), Bayesian Analysis, Time Series Forecasting, Anomaly Detection.
- **MLOps & Deployment:** AWS (SageMaker, Lambda, EKS, S3, Feature Store), GCP (Big Query, Vertex AI), Docker, Kubernetes, MLflow, Kubeflow, CI/CD (Jenkins, GitHub Actions), Flask, FastAPI.
- **Data Engineering:** Apache Spark, Airflow, Kafka, ETL Pipelines.
- **Visualization:** Power BI, Tableau, Matplotlib, Seaborn.

EXPERIENCE

ADP

AI/ML Engineer.

Apr 2025 – Present

- Engineered an LLM-powered HR assistant using GPT-4 + LangChain with RAG (Pinecone/FAISS), implementing NLP models with entity recognition, intent classification, and sentiment analysis to improve query resolution by 30% and reduce escalations by 25%.
- Deployed advanced churn prediction pipelines on SageMaker using predictive modeling techniques, achieving 20% improvement in accuracy for employee retention forecasting and automating retraining with Feature Store + CI/CD workflows.
- Engineered real-time ETL pipelines using Apache Spark and Kafka to process transactional HR and payroll data for enterprise clients, enabling immediate access to employee performance metrics and reducing data retrieval time by 40%.
- Implemented time series forecasting models (ARIMA/Prophet) to predict payroll volumes and processing times, optimizing resource allocation and reducing processing lag times by 15%.
- Developed anomaly detection systems using isolation forests and autoencoders to identify payroll irregularities and fraud, achieving 95% detection accuracy and reducing investigation time by 40%.
- Built a personalized recommender system for training modules based on employee roles, skill gaps, and learning behaviors, contributing to a 25% increase in client engagement with training programs.
- Optimized model performance through advanced hyperparameter tuning in GANs and NLP models, reducing latency by 15% for high-volume HR queries while maintaining peak performance and reliability.
- Implemented explainable AI solutions using SHAP for transparency in payroll and HR decisions, ensuring compliance with ethical AI standards and presenting model insights to stakeholders for regulatory adherence.
- Established robust CI/CD pipelines using Docker, Jenkins, and Git for machine learning model deployment, with comprehensive version control and rollback capabilities.
- Collaborated cross-functionally with engineering, data, and HR teams in Agile environments to refine data pipelines, ensure data quality and integrity, and enhance model deployment timelines and efficiency.

Taylor Corporation

AI/ML Engineer

Feb 2024 - Jan 2025

- Developed GenAI-powered recommender systems using LangChain + RAG (Pinecone) with content-based filtering and NLP techniques (sentiment analysis, topic modeling via NLTK, SpaCy, Gensim), boosting user engagement by 20% through personalized content delivery.
- Architected end-to-end multimodal ML pipelines using PyTorch + PySpark + MLflow with Flask RESTful APIs, streamlining model deployment by 25% and enabling rapid iteration for product teams and stakeholders.
- Built real-time conversational AI systems using Azure Bot Service + LangChain, integrating NLU capabilities with Azure Cognitive Services and RAG models to improve response accuracy by 30% and reduce query resolution time through contextually relevant answers.
- Managed large-scale data processing (trillions of records) using PySpark, and Databricks on AWS/Azure, creating optimized workspaces and clusters that improved data processing efficiency by 15% for marketing insights and strategic planning.
- Automated cloud infrastructure and ETL workflows with AWS Lambda functions, Snowflake integration, and Adobe

Analytics pipelines, reducing operational costs while enhancing data processing efficiency for customer intelligence.

- Implemented comprehensive ML model development using diverse algorithms (Logistic Regression, KNN, Gradient Boosting, Random Forests, XGBoost, SVMs) with ensemble methods, feature engineering, PCA, and hyperparameter tuning to enhance predictive accuracy and model robustness.
- Established robust ML lifecycle management with Kubeflow + CI/CD pipelines (Docker, GitHub Actions), enabling scalable model training, experimentation, and deployment while managing imbalanced datasets using SMOTE techniques.
- Collaborated cross-functionally with product teams in Agile environments, improving project delivery timelines by 20% through streamlined development processes and enhanced decision-making capabilities for marketing strategies

Hyundai Motor India Engineering Pty Ltd

Data Scientist,

Jul 2019–Dec 2022

- Implemented time-series demand forecasting with Python + SQL, improving production planning accuracy by 13%.
- Applied simulation-driven design analytics to evaluate component reliability using past model data and virtual testing, accelerating product development and reducing prototype testing cycles.
- Delivered ML-driven defect detection systems on assembly lines using sensor data, minimizing quality control failures and optimizing inspection workflows.
- Designed predictive maintenance models for manufacturing machinery using IoT data and historical downtime trends, boosting equipment uptime and reducing unplanned stoppages.
- Led integration of sensor fusion pipelines (cameras, vehicle telemetry, ADAS data) to drive next-gen safety and autonomy features.
- Analyzed connected car data and telematics to improve infotainment personalization, remote diagnostics, and customer engagement.
- Collaborated on customer behavior modeling to inform product feature enhancements and targeted marketing campaigns.
- Partnered with cross-functional teams to operationalize predictive models and dashboards, enabling real-time insights for product, engineering, and supply chain

PROJECTS AND ACADEMIC ACHIEVEMENTS

Metaphor Detection (NLP): Trained a DistilBERT-based model to detect metaphors in text, achieving a state-of-the-art NLU loss of 0.045 using Python and TensorFlow.

Crop row and Leaf Segmentation (Deep Learning): Constructed U-Net models achieving 85% accuracy in crop row detection and 97% in leaf segmentation, enabling precise yield assessment and optimized planting.

Click-Through Rate Prediction | Marketing Campaign Optimization: Formulated a click-through rate prediction model on highly imbalanced marketing data (~1.6% positive rate), optimizing for log loss and calibrated probabilities rather than raw classification accuracy, ensuring reliable ranking of high-intent users.

Bayesian Health Insurance Cost Prediction: Constructed a Bayesian model using MCMC for predicting health insurance costs; achieved R-squared of 0.89 and RMSE of 0.46 using Python and R.

Agentic AI for Data Analysis: Developed an autonomous AI agent using LangChain Agents + OpenAI GPT-4 to orchestrate complex analytical workflows (SQL querying, REST API integration, and automated visualization), achieving 95% accuracy in automated insights generation.

LLM from Scratch: Implemented a transformer-based language model from scratch in PyTorch with multi-head attention mechanisms, positional encodings, and layer normalization, achieving perplexity score of 45.2 on text completion tasks through custom tokenization and gradient optimization strategies.

Revenue Forecasting (Time Series): Built ARIMA and Prophet models on 5 years of e-commerce data, achieving 8.5% MAPE for monthly revenue prediction and supporting quarterly financial planning.

Book Recommender System: Designed a hybrid engine on 10M+ interactions using SVD matrix factorization and TF-IDF, achieving 87% precision@10 for personalized book recommendations.

EDUCATION

Purdue University, *Master of Science*

Jan 2023 – Dec 2024 | USA

- Major : Computational Data Science (Computer Science & Mathematics)
- GPA: 3.5/4.0, Dean's Scholarship Recipient

Gayatri Vidya Parishad College of Engineering, *Bachelor of Technology*
India

Jun 2015 – May 2019 |

- Major : Mechanical Engineering
- CGPA: 8.63/10.00, Full-ride scholarship